Task 1000 – Predictive Analytics: Multi-Horizon Forecasting

Tesla, Google, Bitcoin, Ethereum

Yude Li 2025-9-6

Contents

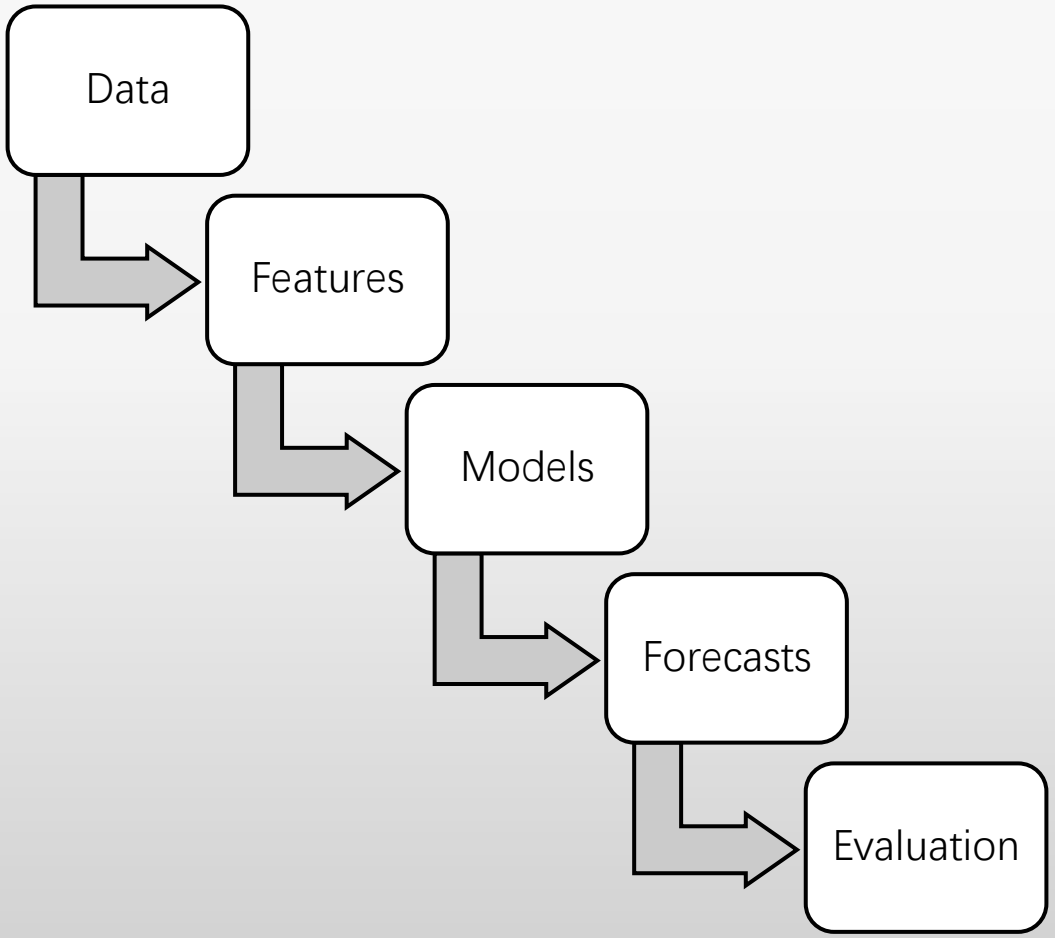
- Overview
- Data Collection & Preparation
- Feature Engineering
- Modeling Approach
- Model Evaluation
- Forecast Results
- Interpretation
- Future Evaluation

Contents

- **Overview**
- Data Collection & Preparation
- Feature Engineering
- Modeling Approach
- Model Evaluation
- Forecast Results
- Interpretation
- Future Evaluation

Overview

- **Objective**:
  - Forecast TSLA, GOOGL, BTC, ETH prices across 7 horizons (30D → 10Y).
- **Horizons**:
  - 30D, 3M, 6M, 1Y, 3Y, 5Y, 10Y
- **Approach**:
  - SARIMAX
  - GBM Monte Carlo



Contents

- Overview
- **Data Collection & Preparation**
- Feature Engineering
- Modeling Approach
- Model Evaluation
- Forecast Results
- Interpretation
- Future Evaluation

Data Collection: Assets & Macros

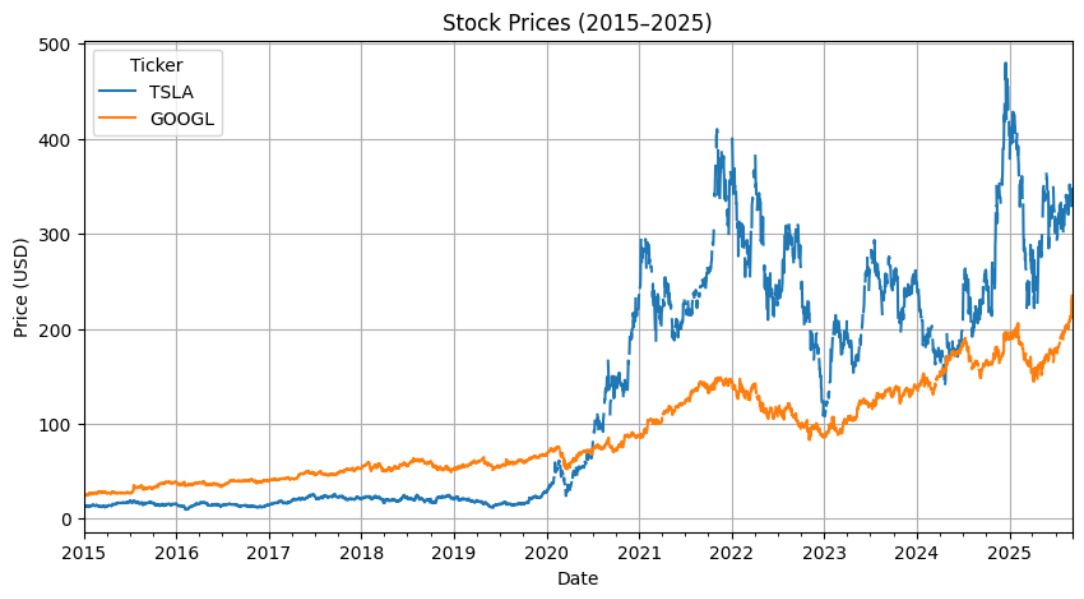
Category	Ticker
Stock	TSLA (Tesla Inc.)
Stock	GOOGL (Google)
Crypto	BTC-USD (Bitcoin)
Crypto	ETH-USD (Ethereum)

Factor	Ticker
SP500	^GSPC (S&P 500 Index)
NASDAQ	^IXIC (NASDAQ Composite Index)
VIX	^VIX (Volatility Index)
US10Y	^TNX (10-Year Treasury Yield)
DOLLAR	UUP (US Dollar Index ETF)
OIL	CL=F (Crude Oil Futures)

- **Source:**
  - Yahoo Finance (2015–2025, daily frequency)
- **Assets:**
  - Key stocks + major cryptocurrencies
- **Macros:**
  - Market indices, volatility, interest rates, currency, energy
- **Purpose:**
  - Provide exogenous variables for forecasting models

Data Collection: Prices (2015–2025)

Historical Stock Prices (TSLA, GOOGL)



Historical Crypto Prices (BTC, ETH)

Data Preparation

- **Frequency Alignment**:
  - Stocks → Business day (B)
  - Cryptos → Calendar day (D)
- **Handling Missing Data**:
  - Forward-fill macro features to align with crypto trading days.
- **Final Dataset**:
  - A unified panel of assets + macro factors, daily resolution.
- **Quick Environment Check**

Ticker	TSLA	BTC-USD	^GSPC	^VIX	…
2025-09-02	329.36	111200.59	6415.54	17.17	…
2025-09-03	334.09	111723.21	6448.26	16.35	…
2025-09-04	338.53	110723.60	6502.08	15.30	…

Contents

- Overview
- Data Collection & Preparation
- **Feature Engineering**
- Modeling Approach
- Model Evaluation
- Forecast Results
- Interpretation
- Future Evaluation

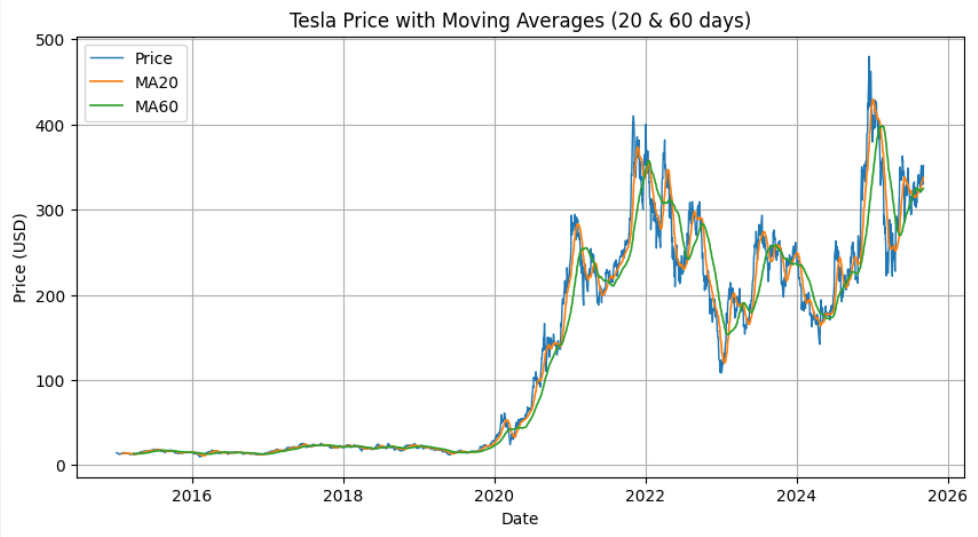
Feature Engineering – Overview

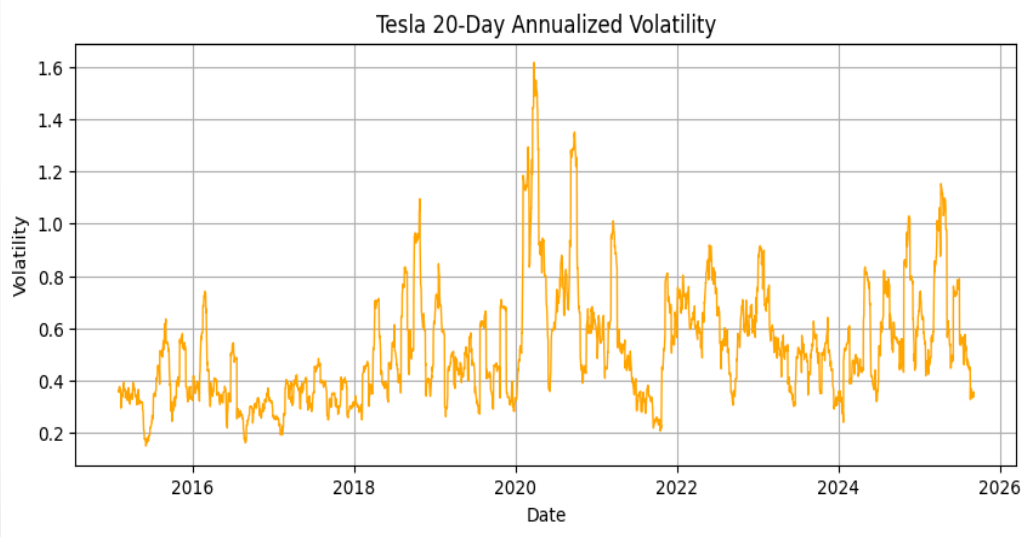
Feature Engineering – Indicators & Factors

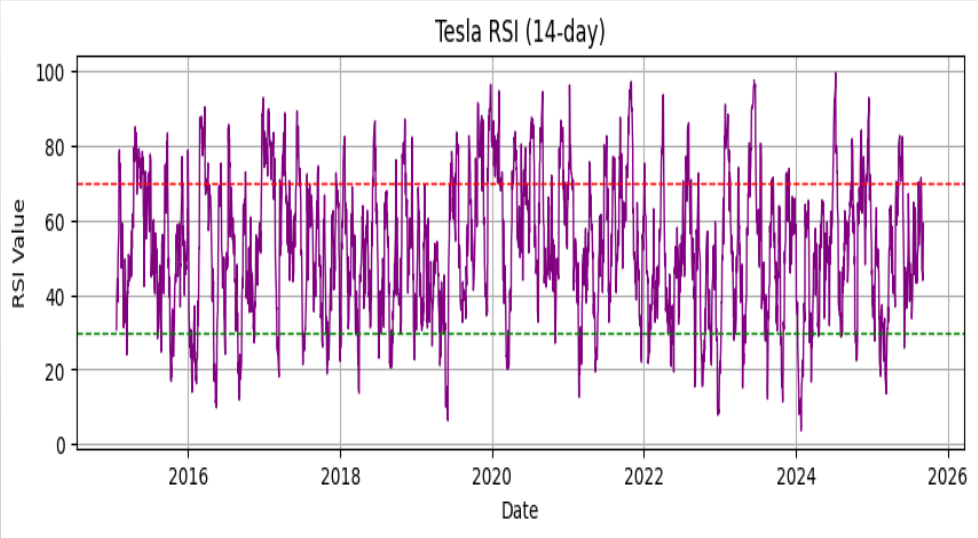
Feature Type	Example(s)	Purpose
Trend	MA(20), MA(60)	Capture long-term price trend
Momentum	RSI(14), MACD	Detect momentum shifts
Volatility	20-day vol	Measure market risk
Macro Factors	VIX_ret, Oil_ret	Reflect external market shocks
Cross-Asset	BTC_ret, GOOGL_ret	Capture interdependence
Lags	lag_ret_5, lag_px_20	Incorporate past dependency

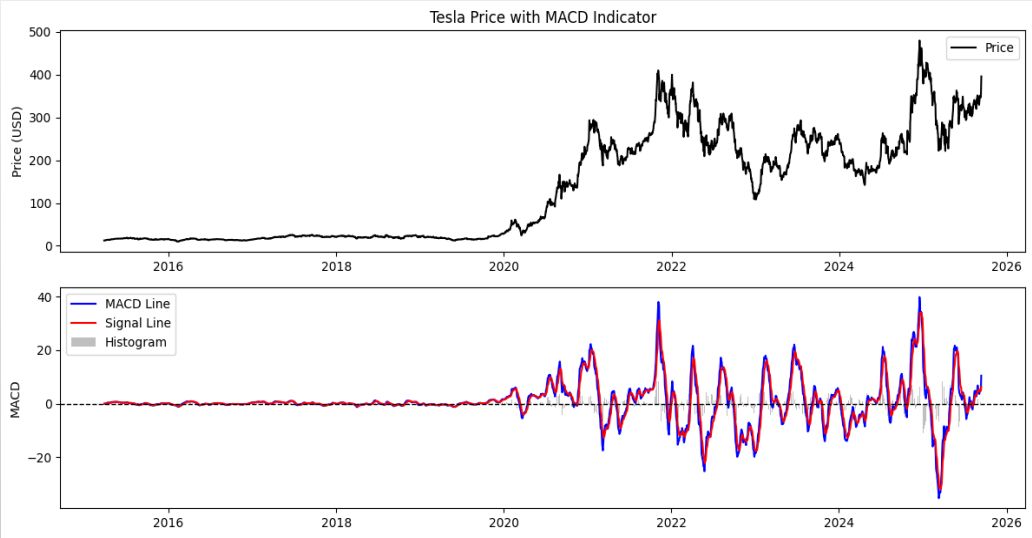
- Technical Indicators: MA(5/20/60), RSI(14), MACD & Signal
- Volatility: 20-day rolling annualized log-return volatility
- Lags: Price/return lags (1, 5, 20)
- Macro Factors: S&P500, NASDAQ, VIX, US10Y, Dollar, Oil
- Cross-Asset Returns: Incorporating TSLA, GOOGL, BTC, ETH dependencies

Technical Indicators — Tesla Example

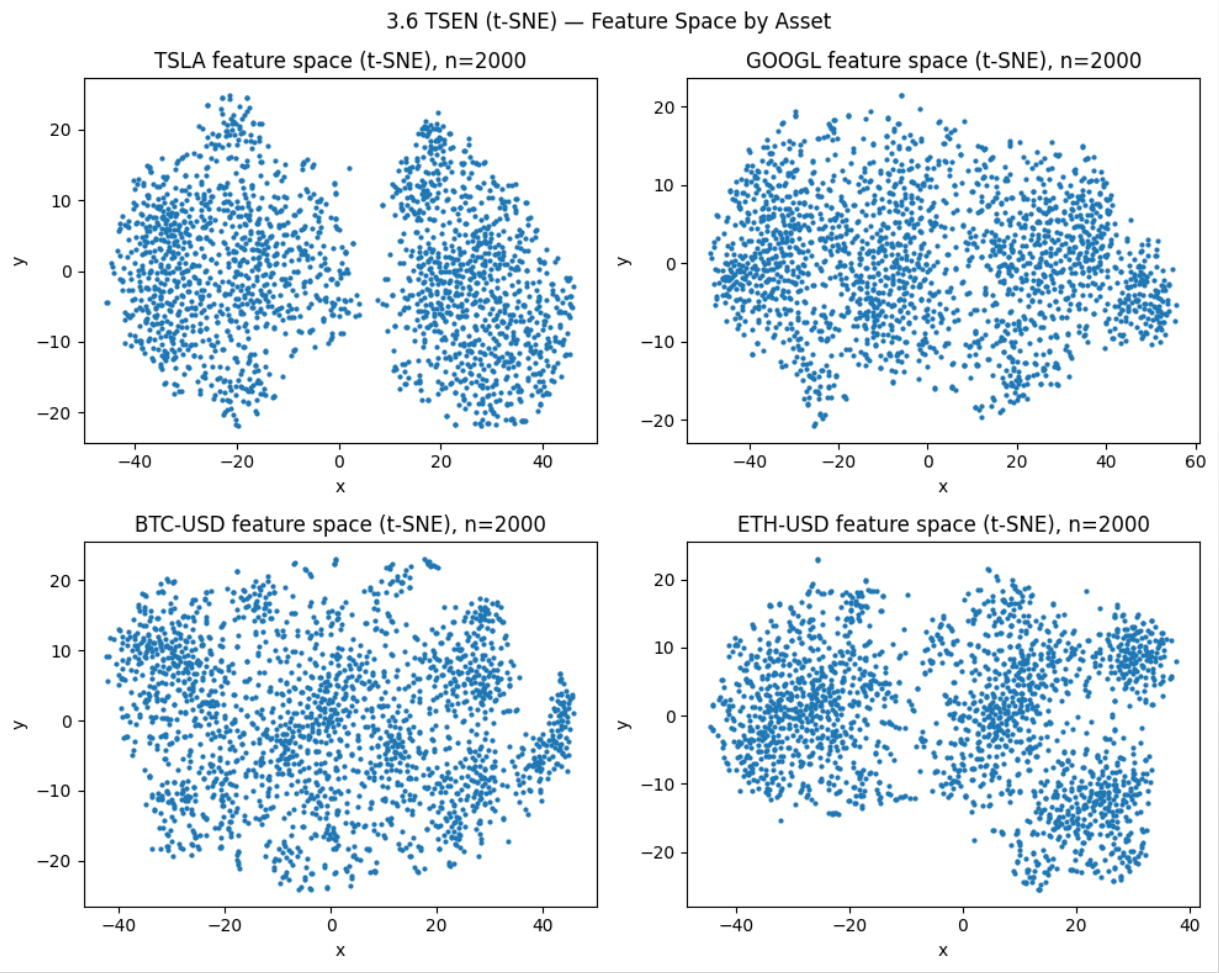








Feature Space Visualization & Summary



- Reduced ~42 engineered features into 2D for visualization, t-SNE shows clusters in feature space
- Suggests potential market regimes
- Cryptos more dispersed, stocks more stable
- Highlights nonlinear structure not visible in raw features

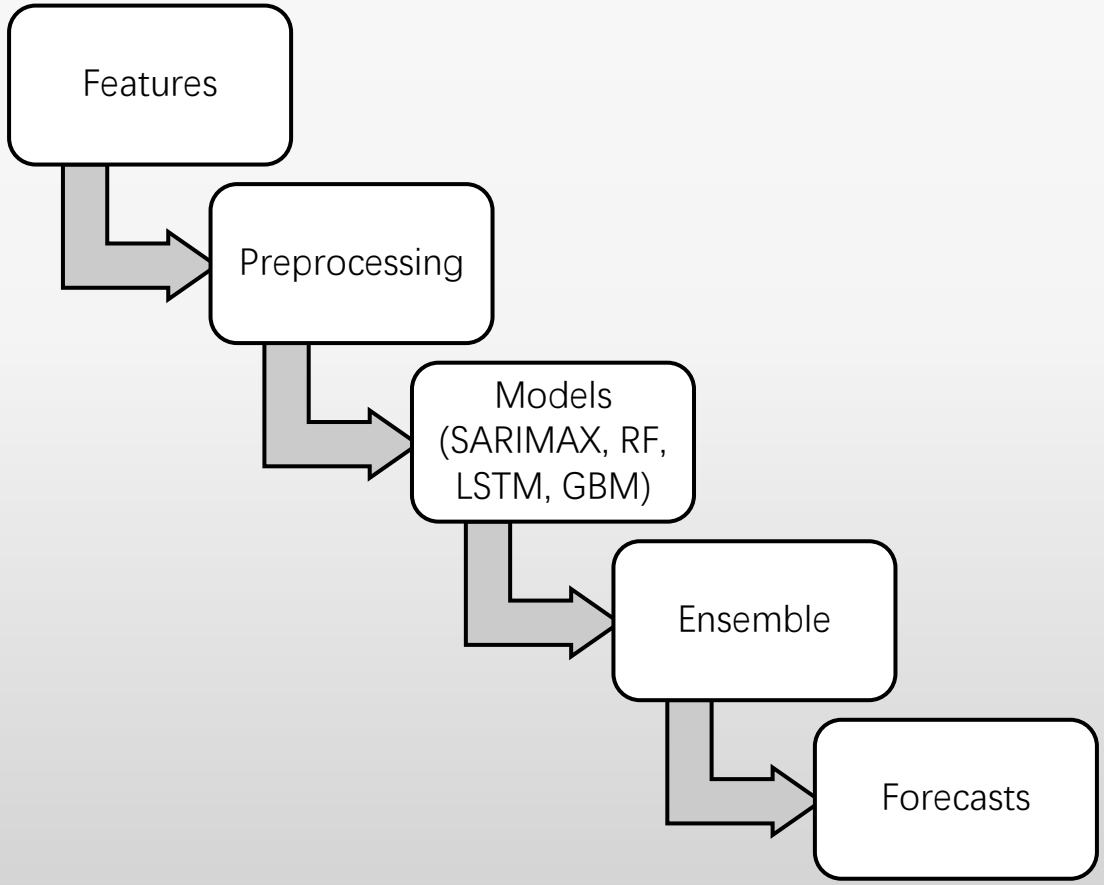
Asset	Rows	Columns	Start	End
TSLA	2725	42	2015-03-27	2025-09-04
GOOGL	2725	42	2015-03-27	2025-09-04
BTC-USD	3840	42	2015-03-02	2025-09-04
ETH-USD	2797	42	2018-01-08	2025-09-04

Contents

- Overview
- Data Collection & Preparation
- Feature Engineering
- **Modeling Approach**
- Model Evaluation
- Forecast Results
- Interpretation
- Future Evaluation

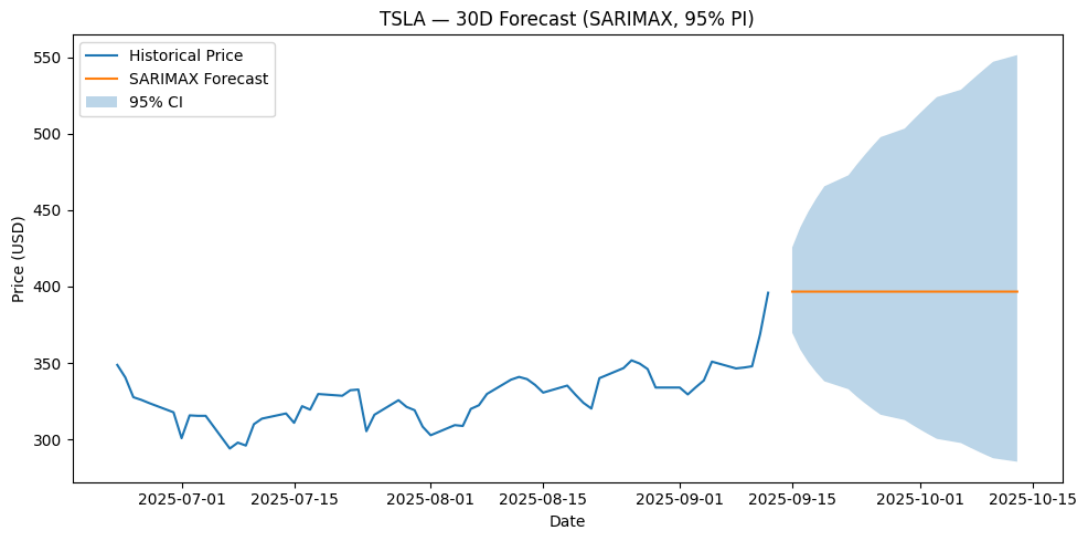
Modeling Approach – Models

- **Short-term (30D–6M):**
  - SARIMAX (with exogenous features), Random Forest, LSTM
- **Long-term (1Y–10Y):**
  - Geometric Brownian Motion (GBM) with GARCH-smoothed volatility
- **Volatility modeling:**
  - GARCH(1,1) with t-distribution, used to refine σ for GBM
- **Ensemble:**
  - Robust median / equal-weight combination for stable forecasts



SARIMAX with Exogenous Features

- Exogenous features:
  - lag-1 versions of macro factors, cross-asset returns, lagged prices/returns
- Preprocessing:
  - StandardScaler + 1%/99% clipping (outlier control)
- Order selection:
  - lowest AIC among (1,1,0), (0,1,1), (1,1,1)
- Steps:
  - 30D ≈ 21 trading steps for stocks, 30 steps for crypto

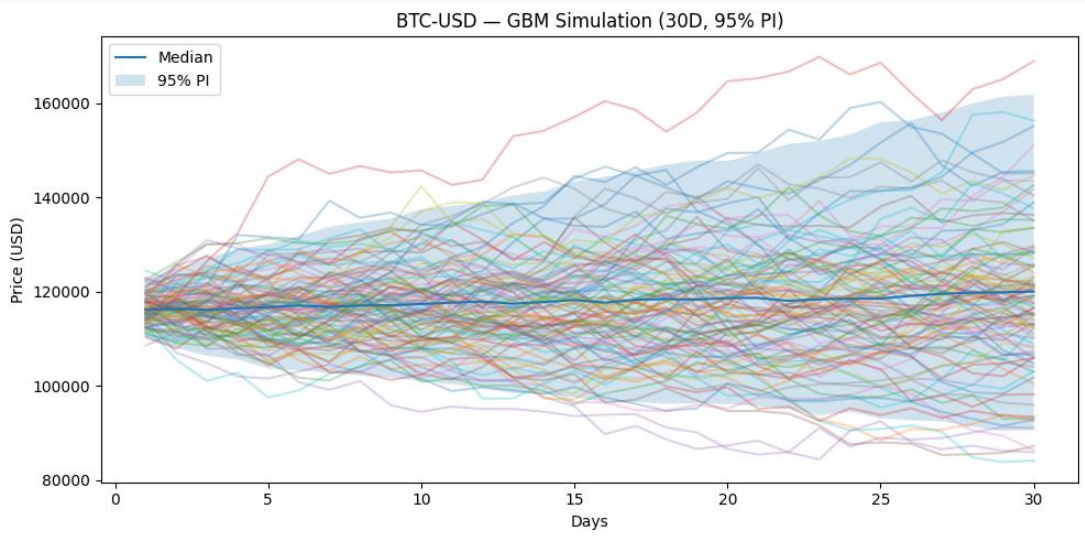


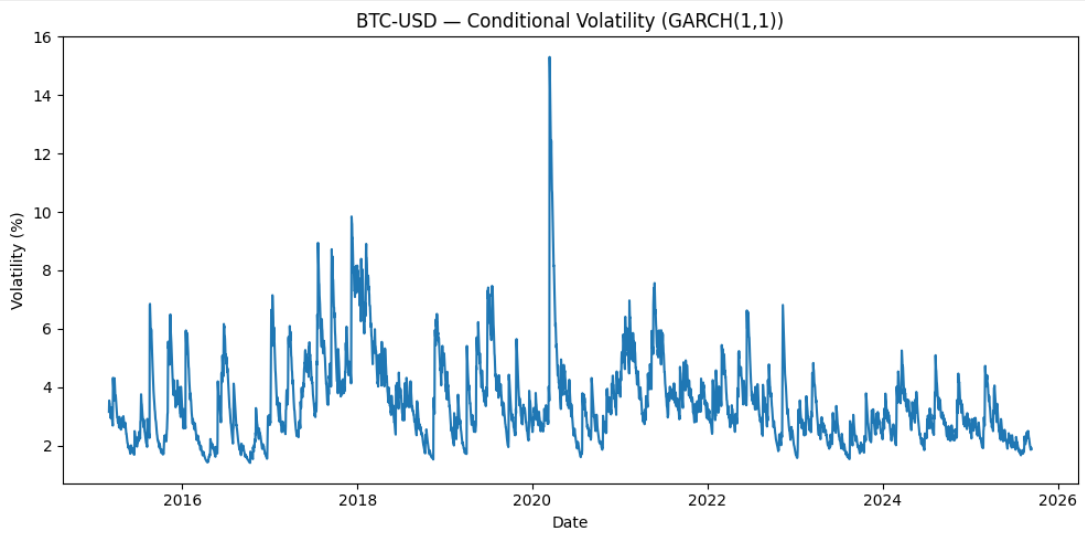
Results (30D, lowest AIC models):

TSLA: (1,1,0), AIC = –10347.34
GOOGL: (1,1,0), AIC = –14185.35
BTC-USD: (1,1,0), AIC = –14760.93
ETH-USD: (1,1,0), AIC = –9355.14

GBM with GARCH-informed Volatility

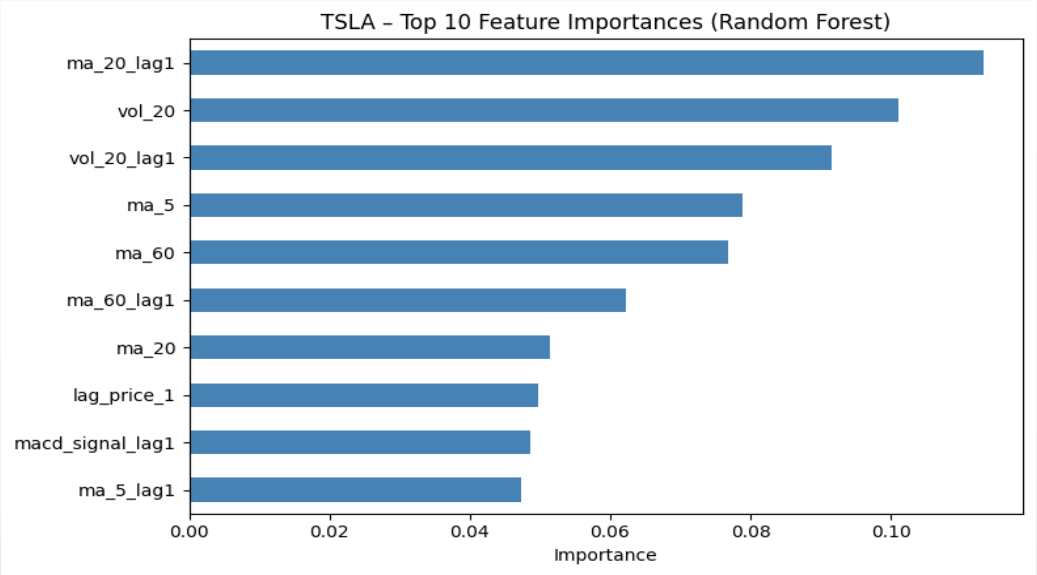
- Parameters:
  - annualized μ and σ estimated from log returns; σ refined by GARCH(1,1) conditional volatility
- Robust constraints:
  - horizon-dependent caps on μ and σ (e.g., μ ⩽ 25% for ⩽1Y, ⩽7% for 10Y)
- Simulation:
  - 2000 paths → median, 2.5%, 97.5% quantiles
- GARCH diagnostics:
  - persistence (α+β ≈ 1)
  - t-distribution ν ≈ 3–4





ML Models & Feature Selection

- **Random Forest (eg: 30D)**
  - TSLA:
  MAE = 0.234, RMSE = 0.287, Forecast ≈ 351.54
  - GOOGL:
  MAE = 0.137, RMSE = 0.153, Forecast ≈ 255.81
  - BTC-USD:
  MAE = 0.348, RMSE = 0.405, Forecast ≈ 106,572
  - ETH-USD:
  MAE = 0.343, RMSE = 0.410, Forecast ≈ 4,590
- **LSTM (eg: 30D)**
  - TSLA:
  val_loss = 0.0346, Forecast ≈ 347.19
  - GOOGL:
  val_loss = 0.00581, Forecast ≈ 230.08
  - BTC-USD:
  val_loss = 0.0144, Forecast ≈ 115,144
  - ETH-USD:
  val_loss = 0.0622, Forecast ≈ 4,334

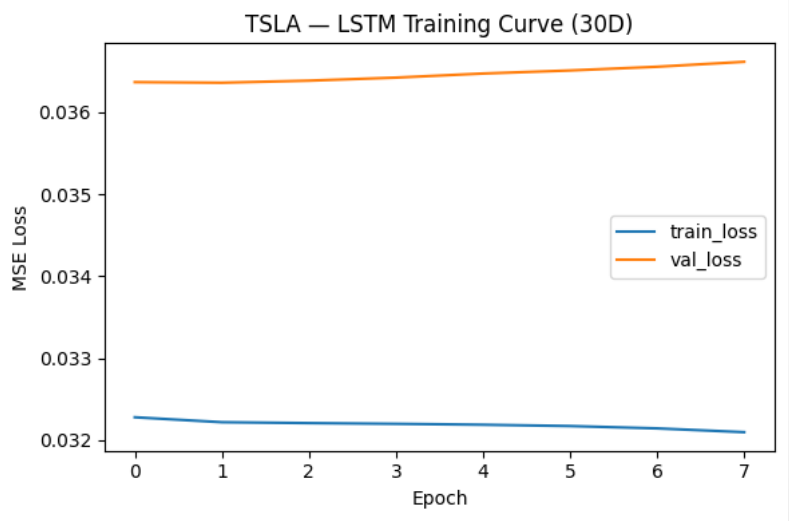


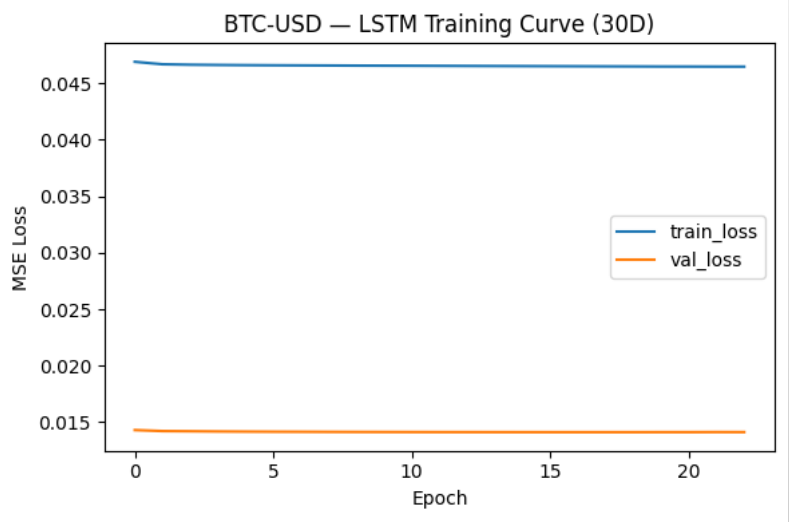
Top Features (Correlation Ranking, eg: 30D)

- TSLA:
  ma20_lag1, ma20, lag_price_5...
- GOOGL:
  rsi14, lag_price_1, ma5...
- BTC-USD:
  ma60_lag1, ma60, lag_price_20...
- ETH-USD:
  lag_price_20, ma20_lag1, ma20...

ML Models - LSTM

- **LSTM Model Design (30D Example)**
  - Input sequence:
  30-day rolling window of daily returns
  - Architecture:
  1 LSTM layer (32 units) + Dense output
  - Training setup:
  Adam optimizer (lr = 0.001), batch size = 64, epochs = 60
  - Regularization:
  Early stopping (patience = 6, restore best weights)
  - Validation:
  90/10 time-based split (no shuffle)
  - Output:
  Point forecasts of log-return sum → converted to price





Ensembling (30D Forecast Examples)

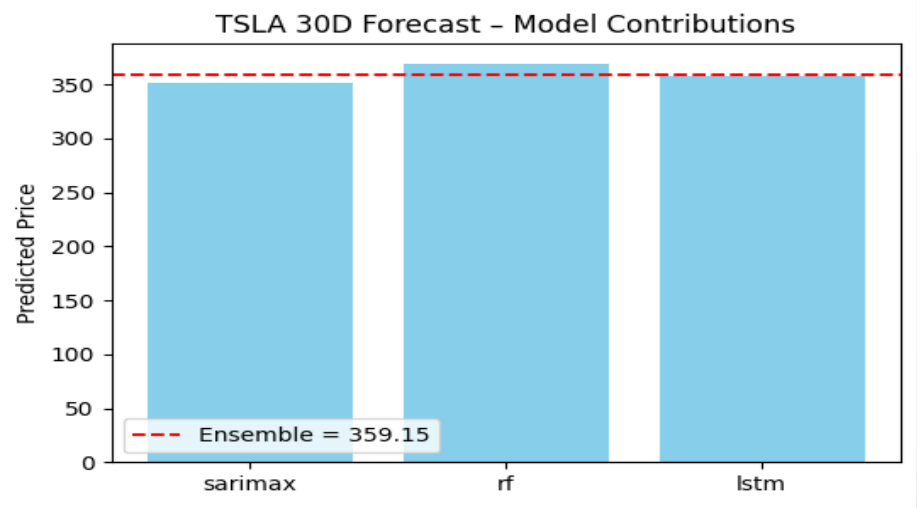
- Robust median across SARIMAX, GBM, RF, and LSTM
- Reduces model-specific bias and improves stability
- Example results (30D):
  - TSLA:

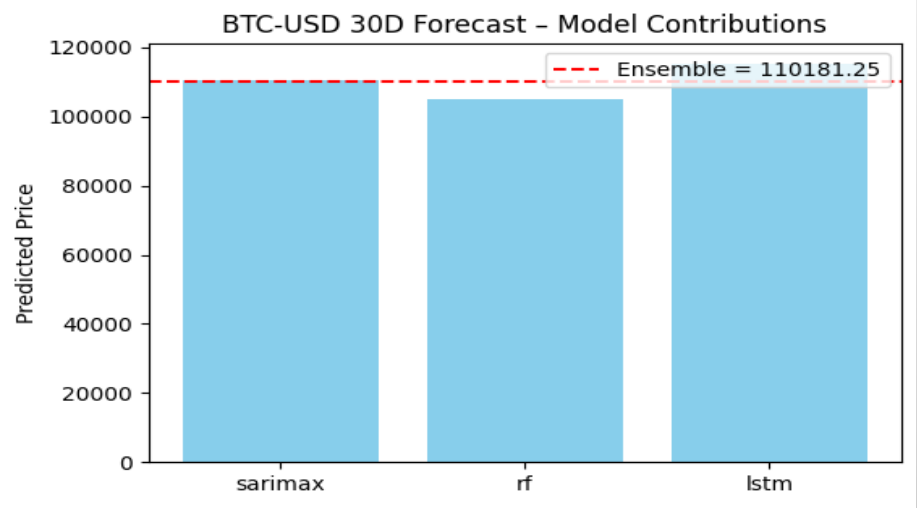
  359.15 (SARIMAX 336.05, RF 351.54, LSTM 347.19)
  - GOOGL:

  237.98 (SARIMAX 228.05, RF 255.81, LSTM 230.08)
  - BTC-USD:

  110,181.25 (SARIMAX 109,564, RF 106,572, LSTM 115,144)
  - ETH-USD:

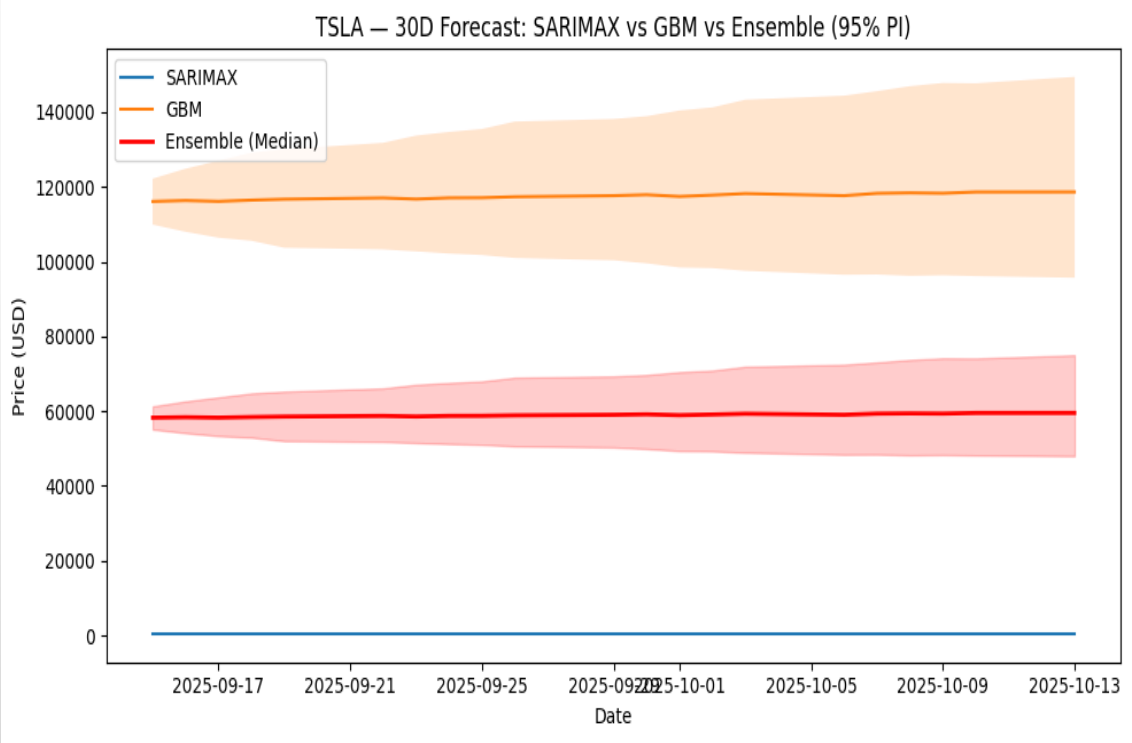
  4,410.71 (SARIMAX 4,308, RF 4,590, LSTM 4,334)





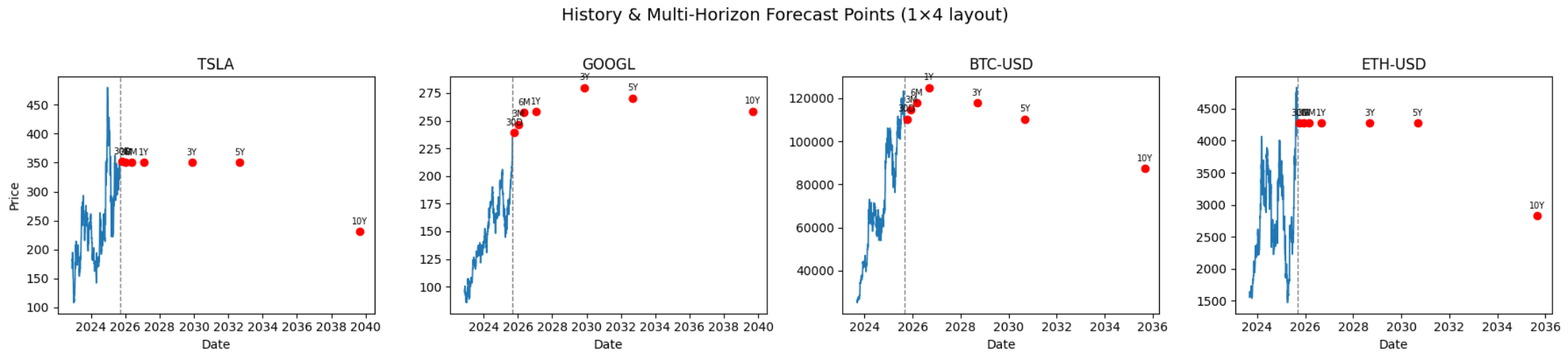
Ensemble Forecasting — Robust Median (TSLA 30D Example)

- **Why Ensemble:**
  Individual models vary across horizons (SARIMAX better short-term, GBM better long-term).
- **Method:**
  Take the **median** of model forecasts; intervals aggregated by median of bounds.
- **Benefit:**
  Stabilizes forecasts, reduces outlier bias, and provides more robust risk bands.



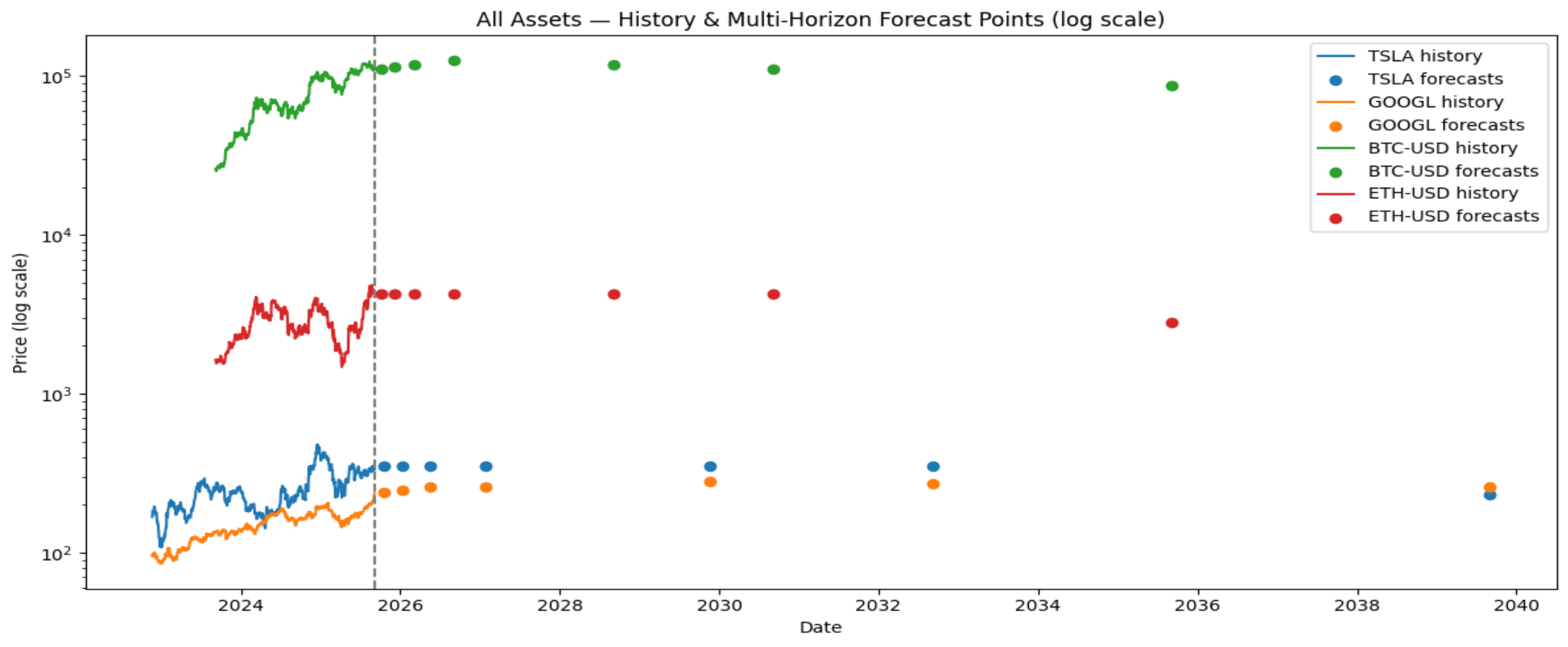
Multi-Horizon Forecasts (30D → 10Y)

Asset	30D	3M	6M	1Y	3Y	5Y	10Y
TSLA	351.72	350.93	350.93	350.93	350.93	350.93	231.18
GOOGL	239.29	246.73	257.26	258.44	279.98	270.53	258.69
BTC-USD	110,308.29	114,621.99	117,754.23	124,938.73	117,765.60	110,308.29	87,325.80
ETH-USD	4,273.63	4,273.63	4,273.63	4,273.63	4,273.63	4,273.63	2,830.82



Multi-Horizon Forecasts (30D → 10Y)

- Log scale is applied to compare assets with very different price levels (stocks vs. crypto).
- Forecast points show ensemble predictions at multiple horizons.



Contents

- Overview
- Data Collection & Preparation
- Feature Engineering
- Modeling Approach
- **Model Evaluation**
- Forecast Results
- Interpretation
- Future Evaluation

Evaluation Framework

Data Splits

- Rolling backtests across 7 horizons: 30D, 3M, 6M, 1Y, 3Y, 5Y, 10Y
- Train/test indices via expanding windows
- Exogenous features scaled (z-score) with 1–99% clipping to reduce leakage/outliers
- Future exog = “last-known snapshot” for SARIMAX; GBM uses μ/σ from recent returns or GARCH
- Models compared: SARIMAX, GBM, RF (direct-H); LSTM used for 30D sanity check

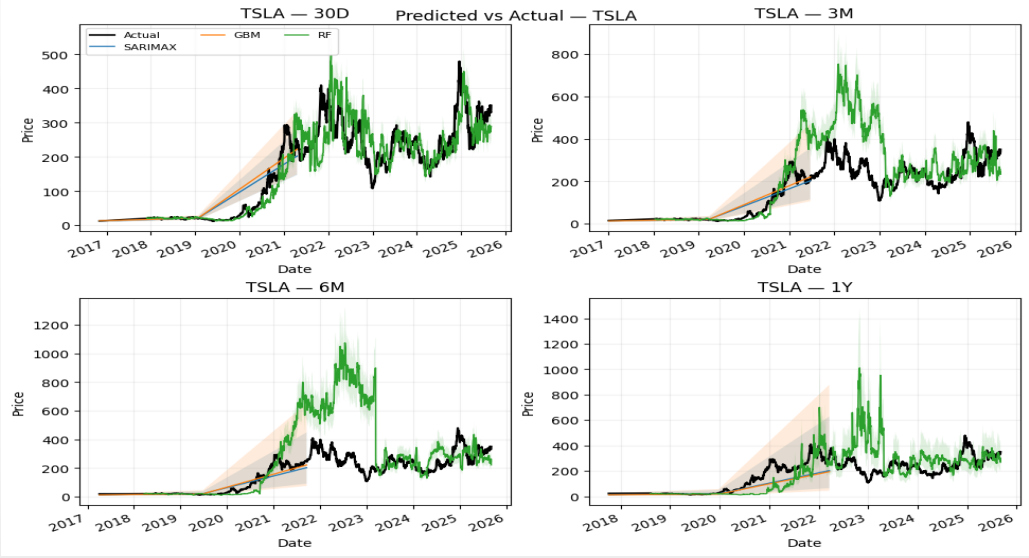
Metrics for Points and Intervals

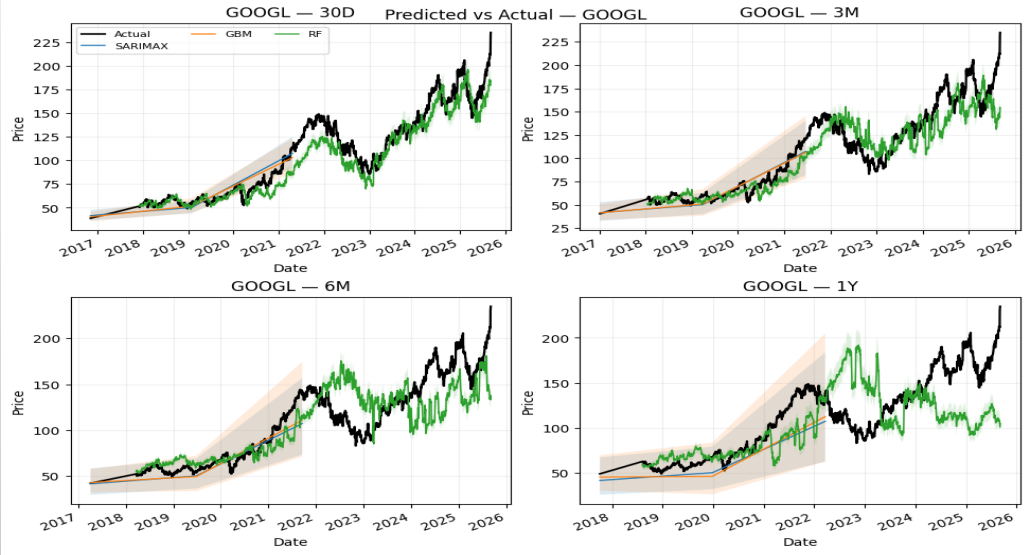
- Point: RMSE, MAE, MAPE, Directional Accuracy (DA)
- Interval: 95% Coverage and Interval Score (proper scoring rule)
- Short horizons prioritize point errors; long horizons emphasize coverage & interval score
- All metrics computed per model × horizon × asset

Backtest Examples (Equities)

Predicted vs. Actual — Equities (TSLA, GOOGL)

- Short-horizon behavior: SARIMAX tracks local mean-reverting moves well
- Medium term: GBM helps when drift dominates; SARIMAX catches cycles with exog
- Intervals illustrate uncertainty growth with horizon

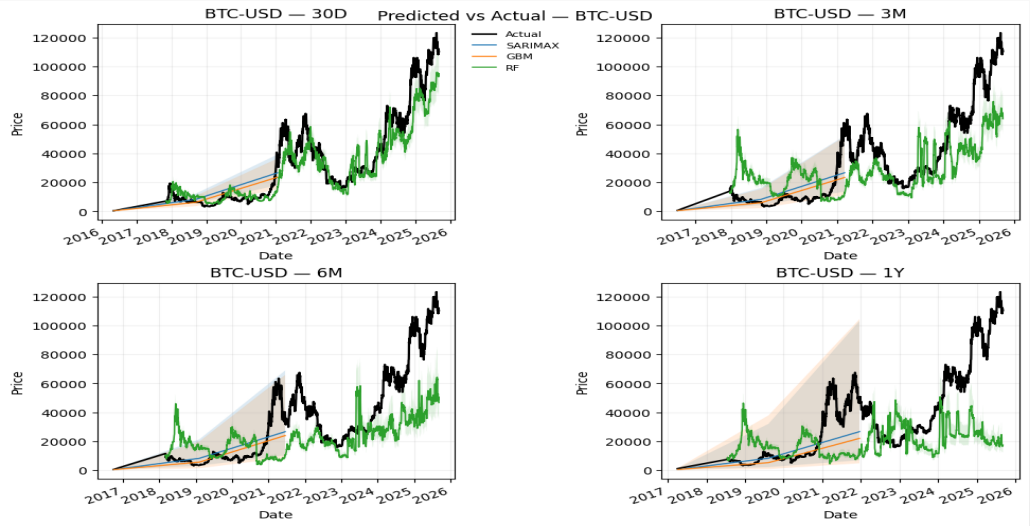


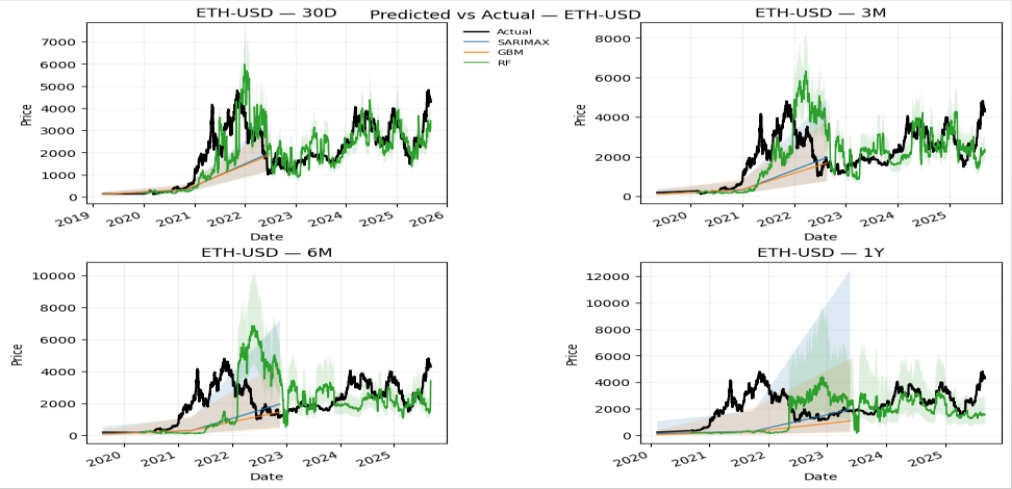


Backtest Examples (Crypto)

Predicted vs. Actual — Crypto (BTC, ETH)

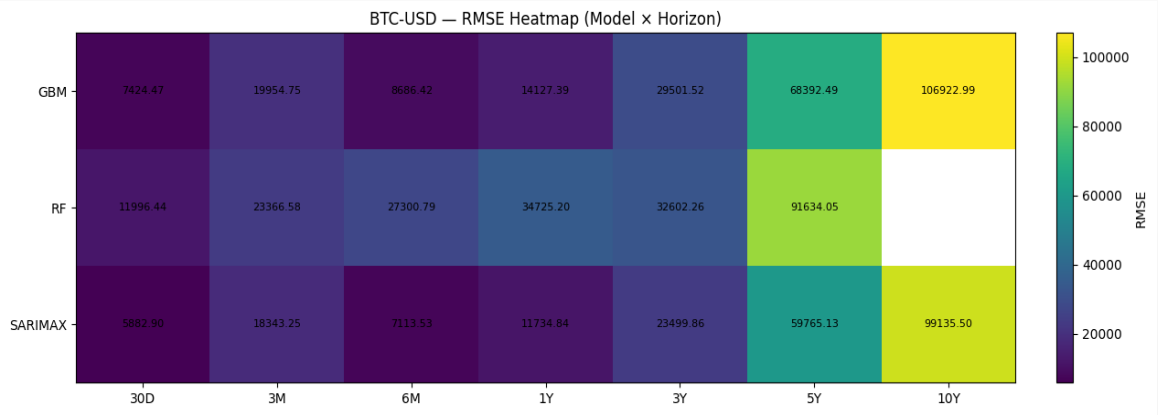
- Higher volatility and weekend trading challenge point forecasts
- GBM captures trend-like moves; SARIMAX helpful when regimes stabilize
- Uncertainty bands widen faster vs. equities

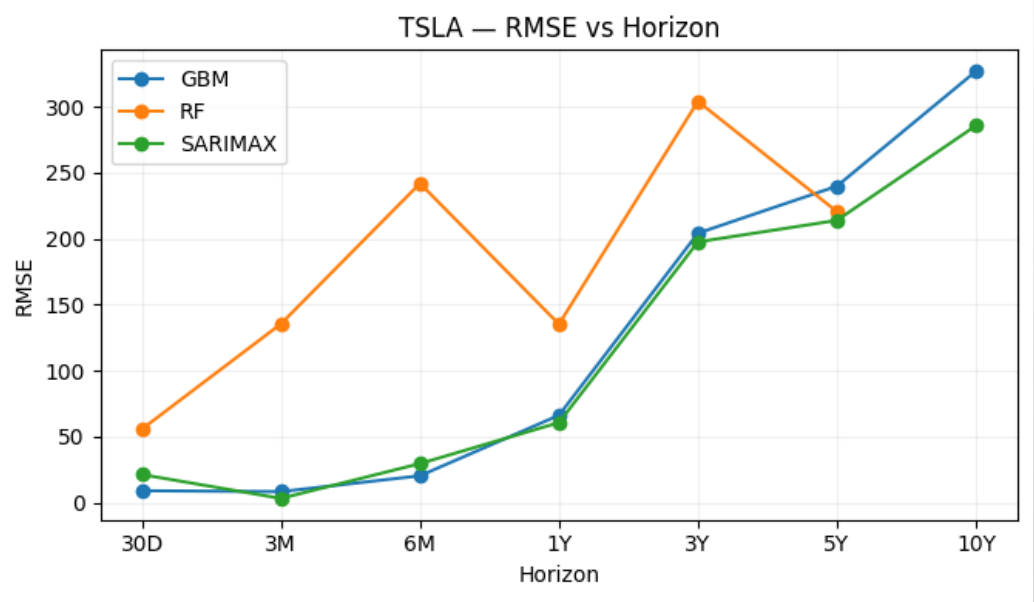




Comparative Errors

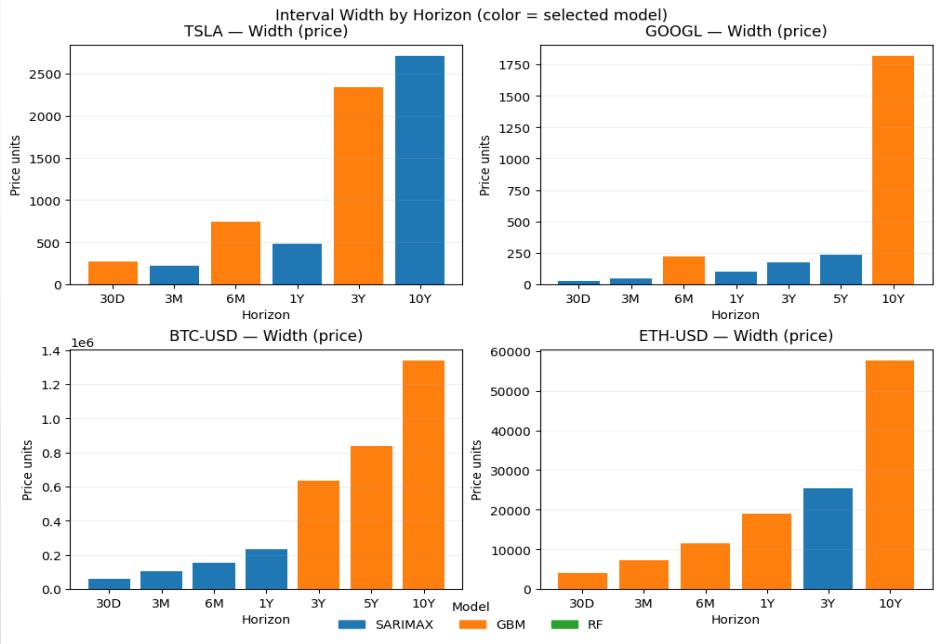
- Short-term (30D–3M): leadership flips — 30D often GBM, 3M tends to SARIMAX
- 1Y: SARIMAX generally most stable (lower RMSE/MAE)
- Long horizons (3Y–10Y): GBM more competitive; RF sporadic wins (e.g., 5Y for TSLA)



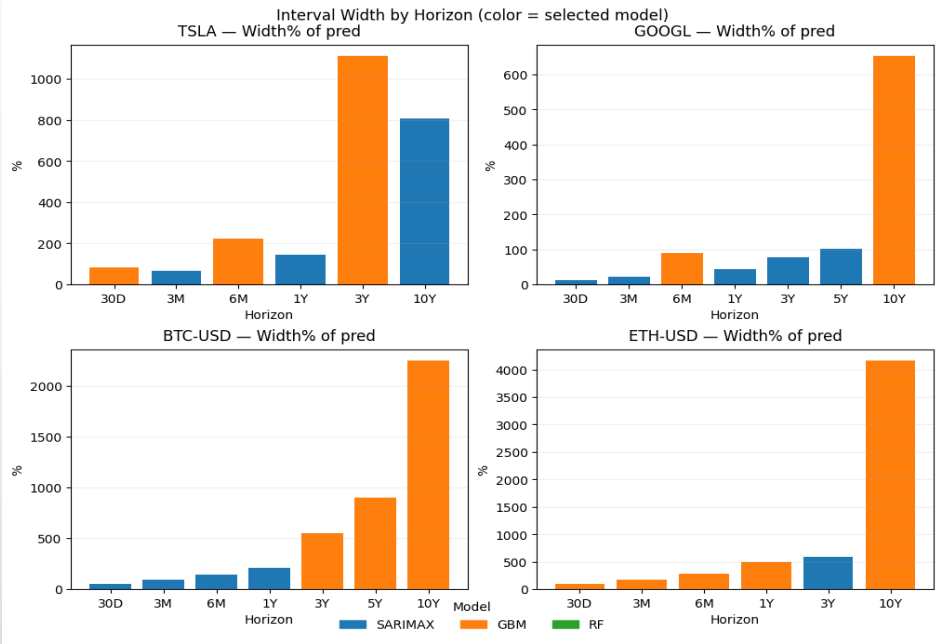


Forecast Interval Width Across Horizons

Interval Width by Horizon (Price units)



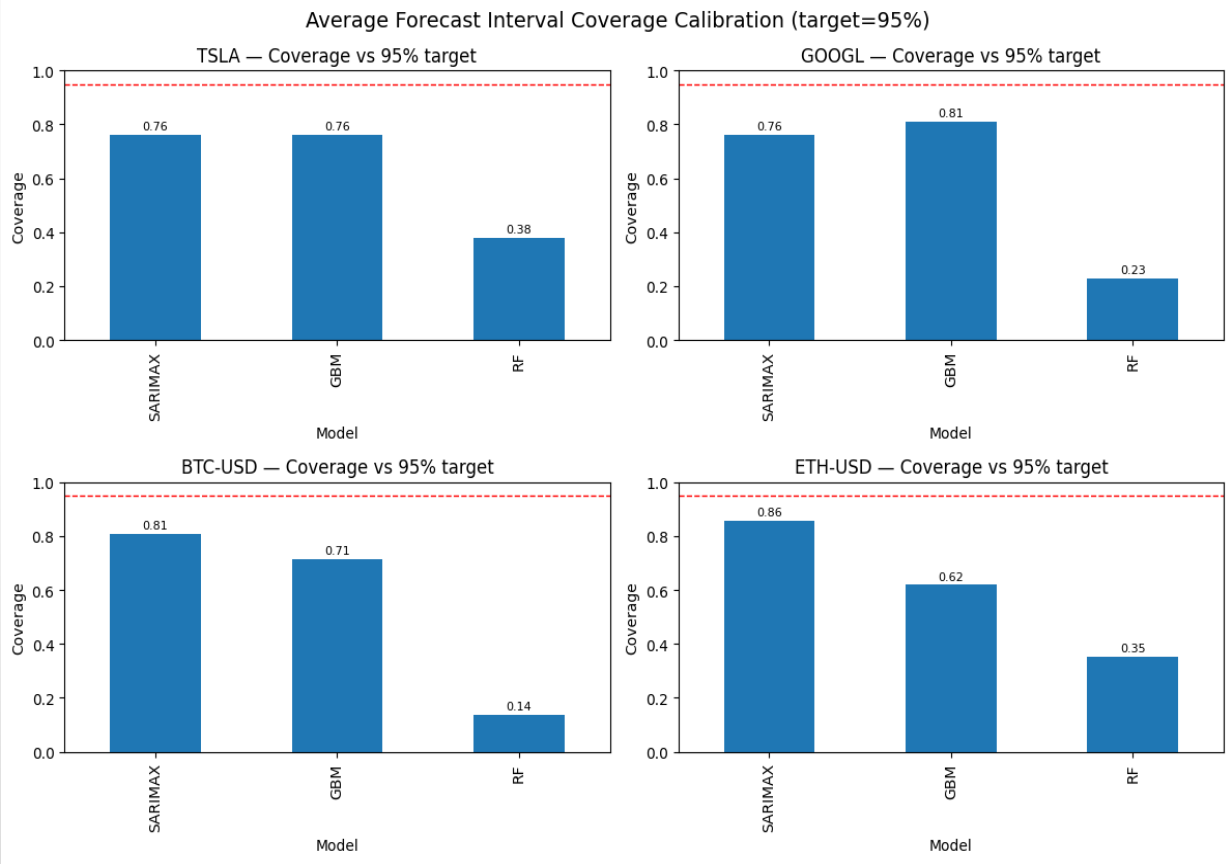
Interval Width by Horizon (% of prediction)



- Forecast intervals naturally widen as horizon increases.
- **Equities (TSLA, GOOGL):** relatively moderate widening.
- **Cryptos (BTC, ETH):** significantly larger intervals → higher long-term uncertainty.
- Percentage view highlights relative uncertainty; price-unit view shows absolute exposure.

Interval Calibration

- SARIMAX intervals are best aligned with 95% target on average
- GBM slightly under/over-covers depending on μ/σ caps
- RF needs conformal/quantile calibration for reliable bands



Best Model by Asset × Horizon

- Heuristic:
  - short horizons prioritize point errors; long horizons prioritize coverage & interval score
- Champions:
  - TSLA: SARIMAX vs GBM trade-off; RF wins at 5Y once
  - GOOGL: SARIMAX dominates across horizons
  - BTC: 30D/3M/6M/1Y → mostly SARIMAX; 10Y → GBM
  - ETH: short–mid → mostly GBM; 5Y RF appears once

	30D	**6M**	**1Y**	**5Y**	**10Y**
TSLA	GBM	GBM	SARIMAX	RF	SARIMAX
GOOGL	SARIMAX	GBM	SARIMAX	SARIMAX	GBM
BTC-USD	SARIMAX	SARIMAX	SARIMAX	GBM	GBM
ETH-USD	GBM	GBM	GBM	RF	GBM

Key Takeaways & Playbook

- **Short/medium term:**
  - SARIMAX is the workhorse; blend GBM under strong drift
- **1Y:**
  - prefer SARIMAX; cross-check with GBM when intervals disagree
- **3Y+:**
  - median ensemble of SARIMAX+GBM balances stability vs. trend
- **RF:**
  - keep as calibrated secondary (direct-H) component

- Report SARIMAX intervals by default; post-adjust GBM coverage; conformal for RF

Contents

- Overview
- Data Collection & Preparation
- Feature Engineering
- Modeling Approach
- Model Evaluation
- **Forecast Results**
- Interpretation
- Future Evaluation

Forecast Results

Intro

- Forward-looking forecasts with best model per asset × horizon
- Short-term (30D–6M) vs Long-term (1Y–10Y)
- Results presented mainly in tables with key insights

Forecasting Framework

- Convert horizons into trading steps (daily vs business days)
- Hold-last scenario for exogenous variables
- Model dispatch: SARIMAX / GBM / RF
- Output: point forecasts + 95% intervals

Short-Term Forecasts (30D–6M)

- **TSLA:** ~335 (GBM/SARIMAX) with widening bands at 6M
- **GOOGL:** ~229 (SARIMAX) stable, 251 at 6M (GBM)
- **BTC-USD:** ~110k (SARIMAX), intervals widen rapidly
- **ETH-USD:** ~4k (GBM), very wide uncertainty

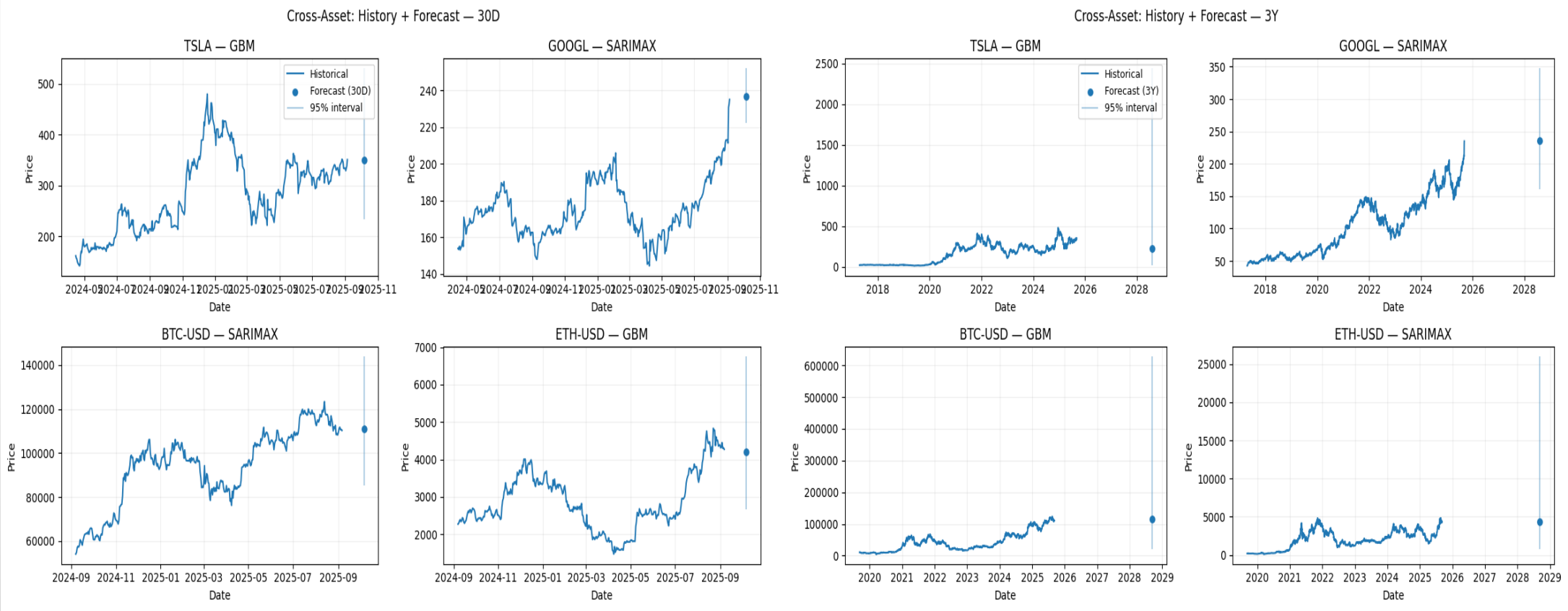
Asset	30D	3M	6M
TSLA	GBM: 350 (234–530)	SARIMAX: 348 (251–483)	GBM: 348 (130–914)
GOOGL	SARIMAX: 237 (222–252)	SARIMAX: 237 (212–264)	GBM: 258 (165–409)
BTC-USD	SARIMAX: 110,821 (85,436–143,749)	SARIMAX: 110,821 (70,201–174,945)	SARIMAX: 110,821 (57,984–211,805)
ETH-USD	GBM: 4,214 (2,675–6,744)	GBM: 4,137 (1,835–9,332)	GBM: 3,942 (1,320–13,365)

Long-Term Forecasts (1Y–10Y)

- **TSLA:** stable medians (~335), wide bands at 10Y
- **GOOGL:** SARIMAX up to 5Y, GBM at 10Y (~278, wide)
- **BTC-USD:** ~110k at 1Y, GBM dominates 3–10Y with extreme ranges
- **ETH-USD:** ~3.9k at 1Y, GBM/RF mix with large uncertainty

Asset	1Y	3Y	5Y	10Y
TSLA	SARIMAX: 348 (179–678)	GBM: 220 (20–2,440)	RF: 844 (point)	SARIMAX: 348 (42–2,865)
GOOGL	SARIMAX: 237 (190–295)	SARIMAX: 237 (161–347)	SARIMAX: 237 (144–388)	GBM: 286 (42–1,997)
BTC-USD	SARIMAX: 110,821 (43,992–279,169)	GBM: 116,427 (20,680–627,325)	GBM: 94,990 (10,970–849,446)	GBM: 62,565 (2,994–1,316,006)
ETH-USD	GBM: 3,897 (815–19,181)	SARIMAX: 4,302 (713–25,957)	RF: 52,032 (point)	GBM: 1,401 (35–55,639)

Cross-Asset Comparison



Compare assets side by side at 30D and 1Y horizons:
Stocks: tighter ranges, more predictable; Crypto: wider ranges, higher risk

Stress Testing — SARIMAX & GBM

- **SARIMAX stress:** macro shocks on exogenous variables (e.g., VIX ↑)

VIX Stress Test (TSLA, 30D)

Scenario	Forecast
VIX +0%	335 (278–404)
VIX +1%	335 (278–404)
VIX +2%	335 (278–404)
VIX +5%	335 (278–404)

- **GBM stress:** scale volatility σ to show uncertainty ballooning

Volatility Stress Test (BTC, 1Y)

σ Multiplier	Forecast
σ ×1.0	122,960 (46,896–324,571)
σ ×1.25	116,083 (35,104–392,377)
σ ×1.5	105,346 (22,995–474,362)
σ ×2.0	85,213 (11,886–590,298)

Equities (SARIMAX) show stable near-term forecasts under macro shocks, while Crypto (GBM) exhibits ballooning uncertainty as volatility rises.

Insights & Takeaways

- **Short-term (30D–6M):**
  - SARIMAX provides tighter, well-calibrated intervals → reliable for equities
  - Crypto forecasts much wider (BTC ~110k, ETH ~4k) → high short-term risk
- **Long-term (1Y–10Y):**
  - GBM dominates; intervals balloon with horizon
  - Equities show moderate widening; crypto uncertainty extreme (e.g., BTC 10Y [3k–1.3M])
  - RF gives point-only signals (not risk-aware)
- **Stress tests:**
  - SARIMAX macro shocks (e.g., VIX ↑) → minimal short-term shifts
  - GBM volatility scaling → median stable but risk bands widen significantly
- **Key takeaway:**
  - Focus on intervals, not just points
  - Blend models: SARIMAX for near-term guidance, GBM for long-term risk envelopes

Contents

- Overview
- Data Collection & Preparation
- Feature Engineering
- Modeling Approach
- Model Evaluation
- Forecast Results
- **Interpretation**
- Future Evaluation

Interpretation

Model Insights

- **SARIMAX:**
  - Best 30D–1Y, interpretable, calibrated; weaker in regime shifts
- **GBM:**
  - Best 3–10Y, captures compounding vol; bands widen fast
- **RF:**
  - Nonlinear short-term signals; point-only, no bands
- **LSTM:**
  - Sequential learner; data-hungry, optional satellite

Short vs Long & Cross-Asset

- **Short-term (30D–6M):**
  - actionable, SARIMAX/RF preferred
- **Long-term (1Y–10Y):**
  - uncertainty dominates, GBM for risk envelopes
- **Stocks:**
  - narrower, macro-sensitive but stable
- **Crypto:**
  - much wider even at 30D; long-term bands extreme

Risk & Takeaways

- Interval widening = main signal; SARIMAX calibrated, GBM conservative
- No model captures black swans → stress testing essential
- **For Investors:** SARIMAX short-term, GBM long-term, re-estimate often
- **For Risk Managers:** run macro (SARIMAX) & vol (GBM) stress; use blended narrative

Contents

- Overview
- Data Collection & Preparation
- Feature Engineering
- Modeling Approach
- Model Evaluation
- Forecast Results
- Interpretation
- **Future Evaluation**

Future Evaluation

Intro

- Forecasting is iterative, not one-time
- Build a feedback loop: Forecast → Validate → Improve → New Forecast
- Ensure models adapt to changing markets

Validation Process

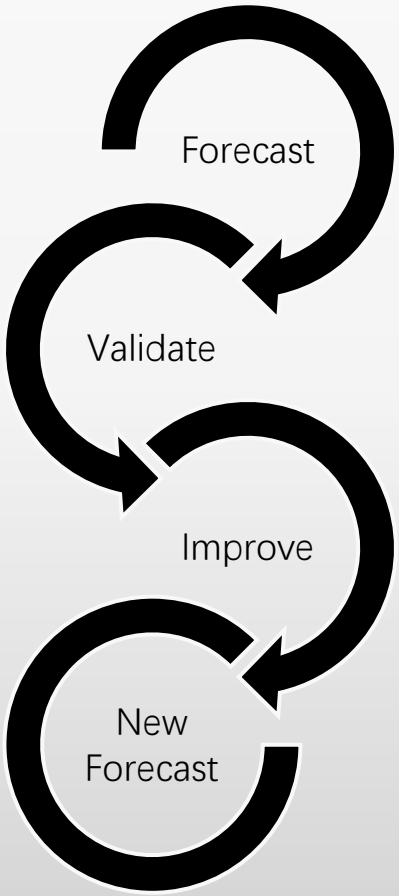
- Compare forecasts with realized prices at horizon endpoints
- Track performance per asset and horizon (30D → 10Y)
- Identify biases, calibration gaps, interval accuracy

Data & Feature Expansion

Category	Examples	Integration
Macroeconomic	CPI, interest rates, unemployment (FRED, OECD, World Bank)	Low-frequency → forward-filled to daily; used as exogenous features
Market Sentiment	Google Trends, news sentiment, Twitter/X indices	Short-term behavioral signals; noisy but timely; merged as predictors
On-chain (Crypto)	Active addresses, exchange flows, hash rate (Glassnode, CryptoQuant)	Provides unique BTC/ETH signals; integrated with price & technical indicators

Closing Note

- Current forecasts = baseline
- Validation reveals strengths & weaknesses
- External features improve robustness
- Continuous cycle keeps framework adaptive



THANKS